Causal Loop and Feedback Analysis

IIIT Hyderabad Breakfast Mess System

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Overview

A causal loop exists only if the last variable in a chain causes the first. This document tests every candidate feedback mechanism in the Breakfast Paradox against that rule. A loop with an even number of negative links reinforces whatever starts it. One with an odd number pushes back toward a steady state. A chain that never closes is neither, so it is drawn as a chain with a dashed closing link.

The evidence is the student interviews, the CFS Chair's account, the CDS poster data, interviews with the operators of the three messes, the team's first-hand account of the Kadamba and Yuktāhār visits, Yuktāhār's photographed registers, and the CDS scan export of every Kadamba breakfast registration in April 2026. Kadamba is run by Prism, Yuktāhār by ABC Hospitality Services, and Bakul by Vijayalakshmi Caterers, which cooks at its Hafizpet kitchen about 8 km away. Whether Palash shares that kitchen is a candidate. Anything read off a register or the scan export, or seen on site, is confirmed. The Yuktāhār operator's statements are confirmed as his account of his own practice. Kadamba and Bakul operator statements are single-sourced: the figures are not in doubt but they are the operators' own claims, and Bakul has no records of any kind.

Eighteen mechanisms are tested, numbered L1 to L17 plus L3b, a reinforcing variant of L3. An earlier analysis had no operator-side evidence and treated the kitchen as a black box. Opening it shows that the best-evidenced feedback in the system lives there.

The Loop Status Dashboard

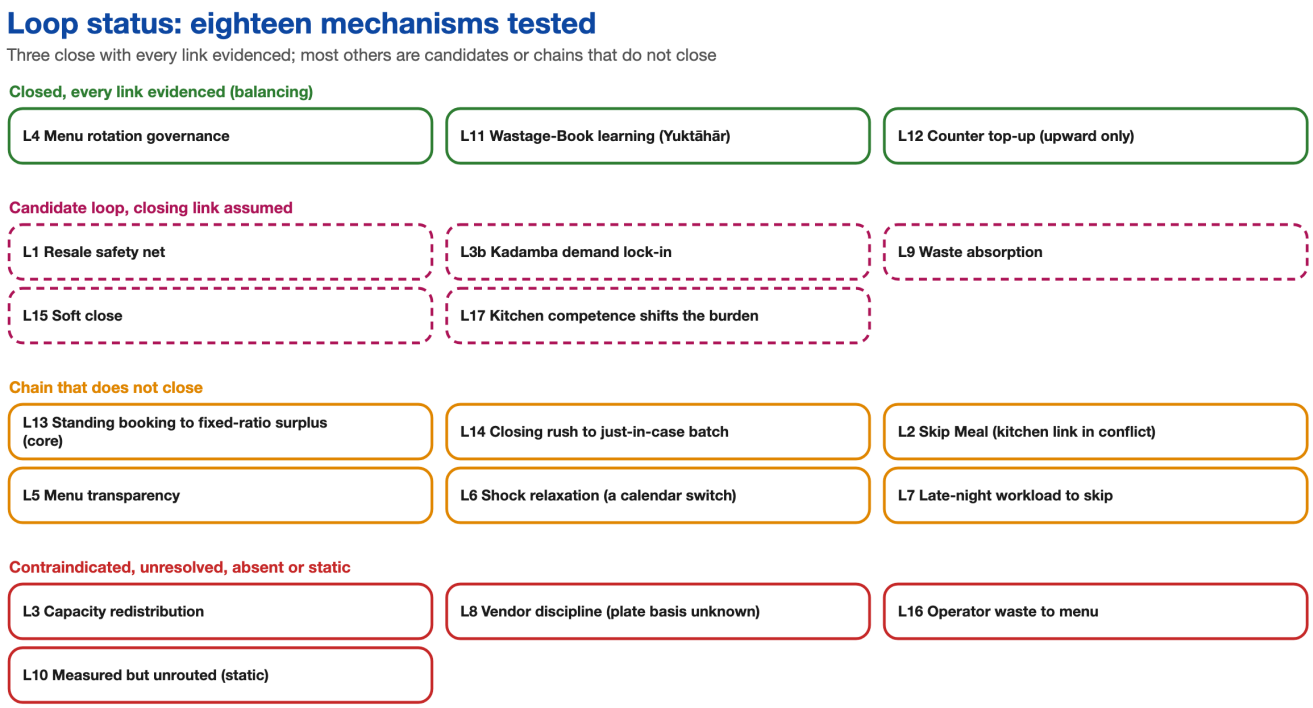


Figure 1. *The eighteen mechanisms by status.*

ID	Mechanism	Status	Delay
L1	Speculative registration and resale	Reinforcing, candidate	Cancel 4 days ahead
L2	Skip Meal	Chain; kitchen link in conflict	Plan the day before
L3	Capacity redistribution	Strongly contraindicated	None found
L3b	Kadamba demand lock-in	Reinforcing, candidate	Registration cycle
L4	Menu rotation governance	Balancing, closes, confirmed	Semester to policy
L5	Menu transparency	Chain	Menu posted 2 weeks ahead
L6	Shock-adaptive relaxation	Chain; a calendar switch	Length of the shock
L7	Late-night workload to skip	Chain; first link confirmed	Overnight
L8	Vendor quality discipline	Unresolved; depends on plate basis	Monthly payment
L9	Waste absorption	Reinforcing, candidate	Cooked food lasts 3 to 4 hours
L10	Measured-but-unrouted gap	Static condition	No route
L11	Operator waste-learning	Balancing, closes, confirmed at Yuktāhār; absent at Kadamba; candidate at Bakul	Next-day plan; 1 to 2 week lookup
L12	Counter top-up	Balancing, closes, confirmed; upward only	10 to 15 minutes on site
L13	Standing booking to fixed-ratio surplus	Chain, the core; open quality branch	4 days, 1 day, 5:30
L14	Closing rush to just-in-case batch	Chain; candidate ratchet	Last 10 to 15 minutes
L15	Soft close	Reinforcing, candidate	Unknown
L16	Operator waste to menu choice	Absent	No route
L17	Kitchen competence shifts the burden	Candidate	Daily

Three close with every link evidenced: L4, L11 and L12. Five are candidate loops (L1, L3b, L9, L15, L17), six are evidenced chains that stop short (L13, L14, L2, L5, L6, L7), three are disconfirmed, absent or unresolved (L3, L8, L16), and one, L10, is a static condition. The split is the finding. The gap is measured and acted on every day inside the kitchen by loops that work, but no route carries those measurements to whoever sets the registration rule. Every working loop corrects the kitchen's conversion of bookings into food, and none touches the booking.

Reinforcing and Balancing Feedback Loops

L11: Operator Waste-Learning (Yuktāhār)

At Yuktāhār, cooked quantity above what is eaten raises the leftover and run-out time written in the Wastage Book. A planning meeting held twice a day reads that record against the matching past day (the menu repeats with weeks one and three identical and weeks two and four identical, so a week-three Thursday is planned from the week-one Thursday), the month's turnout pattern, any events, and registrations counted as regular plus Jain plus cash meals. Its output is the next day's issue list. A larger leftover lowers the next issue and a lower issue lowers the surplus. One negative link makes the loop balancing, and every link is evidenced. The photographed pages record per item the raw quantity issued, cooked, left, percentage wasted, run-out time and re-cook, and the site head confirms their use in planning. The rule has a stated form: ”If it runs out at 1:30 we re-cook; if at 2:20 we don't.” Shown the 18.7% attendance of Sunday 6 September, he said the quantity would be cut ”not even next month, next week.”

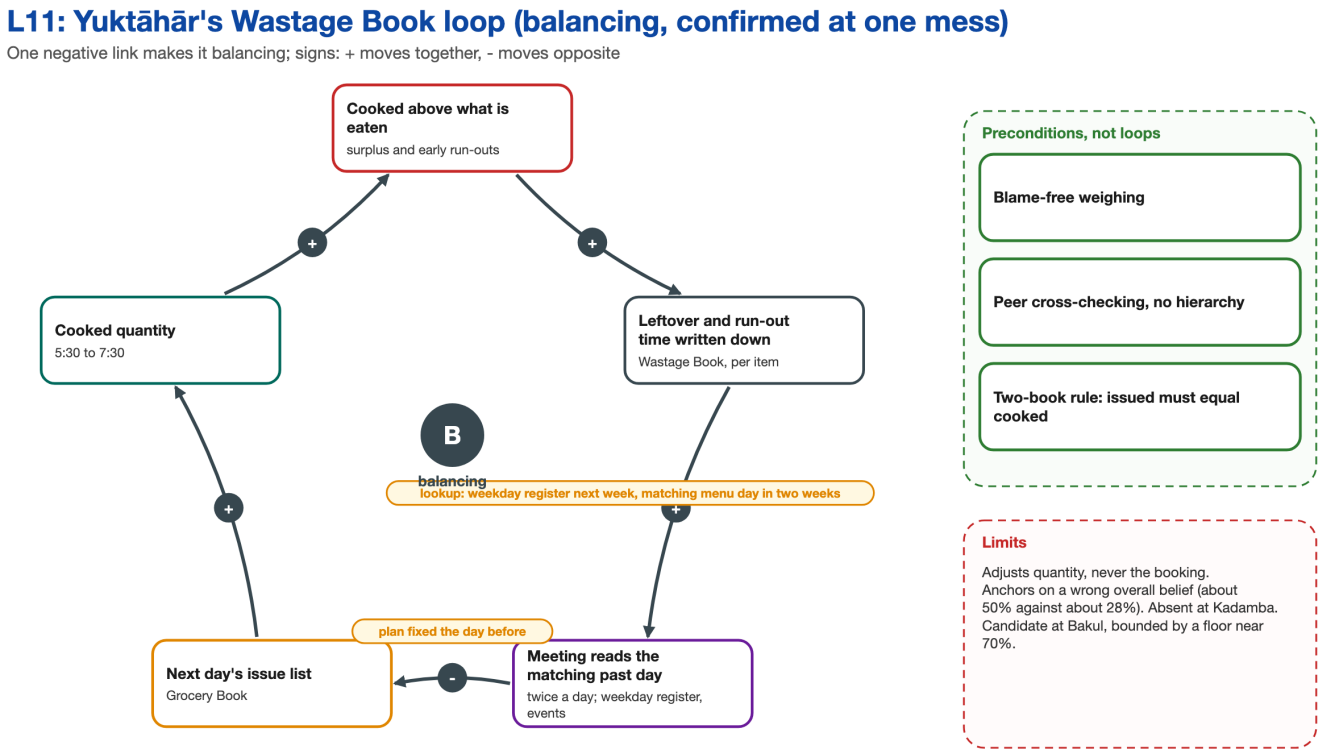


Figure 2. *L11: Yuktāhār's Wastage Book loop, its two delays, and the preconditions that keep the record honest.*

The loop has a goal. The coordinator keeps a wastage board and works to push the waste share down, and the operator names lower cost among its aims, so waste is a cost lever for this vendor. Two delays apply, the plan fixed the day before and a learning lag of one to two weeks. A slower one sits behind the loop's existence. Weighing began in August, staff first feared blame (”they are catching us”), the managers made it blame-free and discussed it daily, and by December staff themselves reported lower waste (self-report; no before-and-after figures seen). Three conditions keep the record honest, and they are preconditions, not loops: blame-free weighing, peer cross-checking by staff who rotate across every job, and a two-book rule that what the store book says was issued must match what the physical book says was cooked. The rule held on 18 September, when 4.5 kg of toor dal and a balance of 49.45 kg agree in the Grocery Book and the stock register (confirmed). It all runs on paper, since the operator tried a commercial ERP system and dropped it as too costly for data already in its books (single-sourced).

Four limits apply. The site head believes breakfast turnout is about 50% while his own register shows 17.7% to 40.2%, about 28% on the legible rows, so the loop corrects item by item while anchoring on a wrong overall figure. It adjusts quantity only, never registration. It rests on one operator's account from one visit, and it is confirmed at

one mess of three. Kadamba keeps per-dish gram standards (four idlis from about 120 g of batter, 100 g of sambar) but no record that learns from turnout: ”We don't have that idea… it was wasted many times.” The three operators work under the same registration rule and hold three stances. Yuktāhār treats waste as lost cost to be learned away, Kadamba cooks on registrations ”because it is a responsibility and they are getting paid for it”, and Bakul's rule is ”There should not be a shortage of food, even if some wastage occurs.” No contract term behind any stance has been seen, so operator culture is a candidate explanation for why the loop is present at one hall, absent at another and bounded at the third.

Bakul is a weaker, bounded candidate second instance. Its manager sets the first batch at 70 to 80% of registrations by item from weekly attendance history (if Mondays have run near 70%, he cooks for about 80%), so a surplus can lower next week's share. The loop closes against surplus only down to a floor: even on weekends, when he calls breakfast attendance very low, he keeps food for about 70% of registrations. Whether the review is written down is unknown, so the measurement may live in one person's head. His estimate of at most about 60% turnout has no record behind it, and on his own numbers the first batch and the floor already sit above it.

L12: Counter Monitoring and Top-Up

Within a meal, food at the counter falls, a periodic check registers the risk of running out, the counter asks for more, and food rises again. One negative link makes the loop balancing, and all three operators describe it.

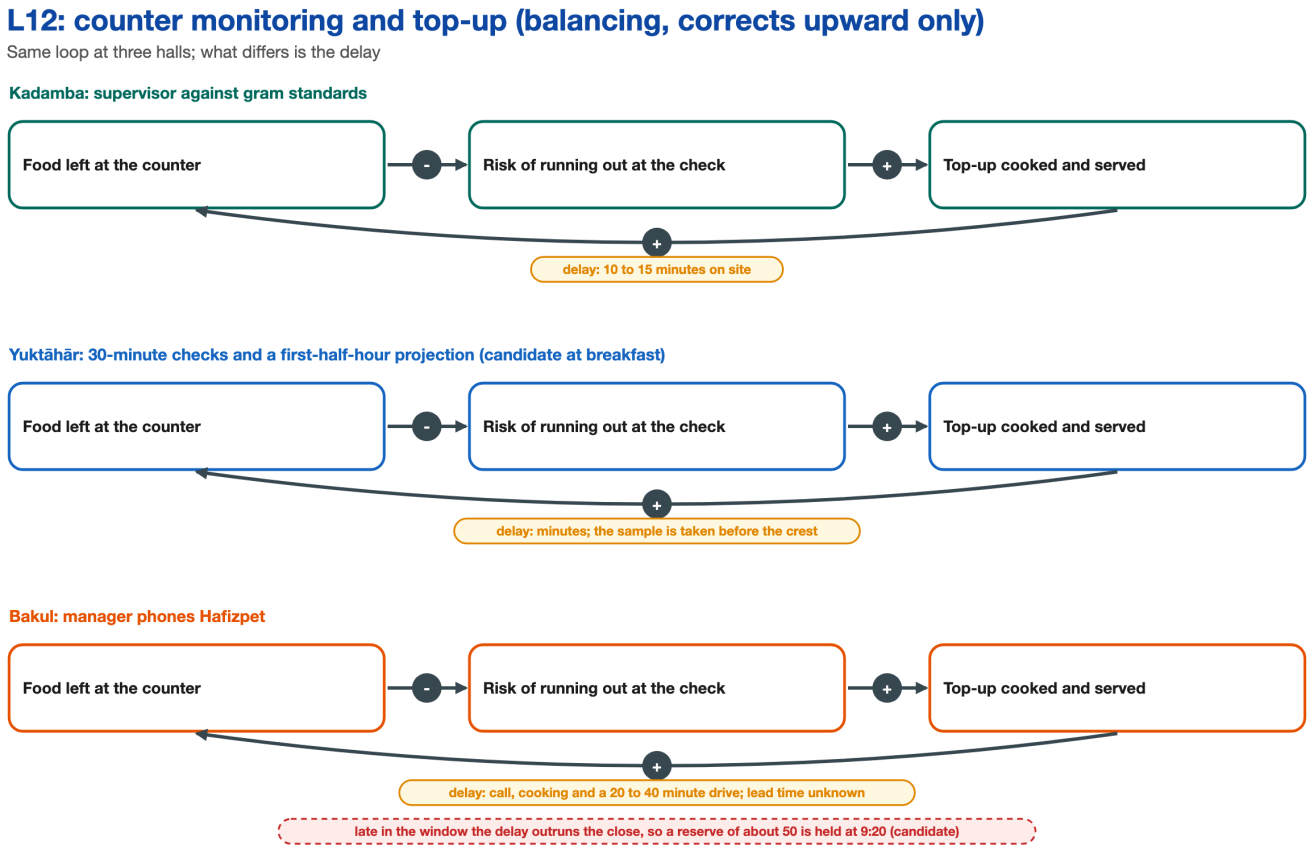


Figure 3. *L12: top-up at three halls, and the delay that decides how well each works.*

At Kadamba a supervisor watches each item's level against its gram standard and asks the chef for a number of portions, combining registrations with turnout so far; the raw remainder is held ready, and top-ups are ready in 10 to 15 minutes (single-sourced). At Yuktāhār staff check every 30 minutes from about 45 minutes before close, open a second counter at surges and re-cook by the run-out rule, and the 9:00 re-cook on 3 September is on record (confirmed). Yuktāhār also runs a forward-looking second form: the first half hour of service is a sample that projects

the rest, which drives top-up cooking (its use at breakfast is a candidate). The weakness is the sample, taken from the part of the morning least like the close. On Kadamba's April curve 15.3% of scans fall between 7:30 and 8:00, so a flat projection of that rate gives about 240 of the roughly 398 daily scans, about 40% short (candidate arithmetic; Yuktāhār's own curve is unmeasured). At Bakul the manager checks about every 30 minutes and phones Hafizpet with a rough quantity, which is cooked and driven over: 20 minutes early in the morning, 40 in normal traffic, total lead time unstated (single-sourced).

The delay decides the behaviour. On site, a top-up requested when the final surge becomes visible lands at or after the 9:30 close, so the loop cannot track the crest and can only be pre-empted by a buffer cooked in advance, which is L14's hedge. Off site the delay is longer, and a call, a cook and a 20 to 40 minute drive cannot land late in the window. The only hedge against late arrivals is food already held, which is why the Bakul manager aims to have food for about 50 people at 9:20 against the 10 to 15 he says arrive in the last ten minutes. As a candidate reading, that reserve is the structural consequence of the transport delay, an insurance premium paid in surplus. Bakul is the clearest case of a balancing loop whose delay is long enough that the kitchen stops relying on it at the end of the meal.

The loop is also asymmetric. Cooked food cannot be un-cooked and lasts three to four hours, so L12 corrects only upward. Surplus inside the meal is carried forward, held and reheated at Kadamba (the open branch of L13), or held hot at Bakul in insulated boxes at about 60 to 72 °C, where the reserve for the latest diners is also the food held longest and likely the softest (candidate). L12 protects against shortage and cannot reduce surplus. That job falls to the slower L11 where it exists, and otherwise to nothing.

L4: Menu Rotation Governance Response

This is the governance loop in which every link traces to a direct account. A single fixed menu for the whole semester produced menu fatigue. The feedback was intense enough to reach the CDS Student Council, which escalated it to the Student Parliament, and the CDS Committee approved a biweekly rotation. Variety rose and, by the CFS Chair's account, fatigue complaints fell. The two-week cycle is confirmed in the operators' records.

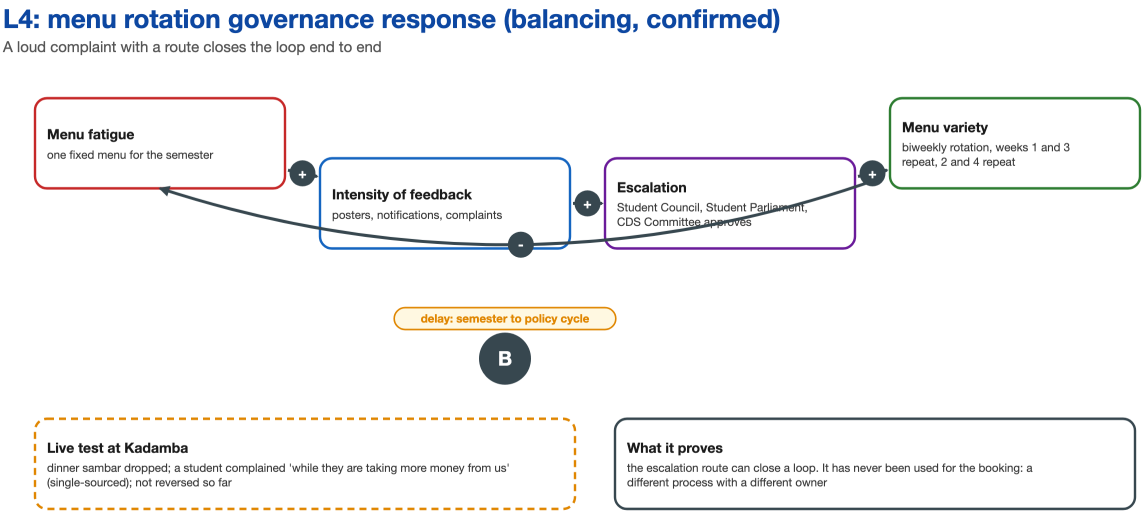


Figure 4. *L4: the menu-rotation loop, closed end to end by a loud complaint with a route.*

The loop shows that the escalation structure can close a feedback loop when a complaint is loud enough, which makes the gaps elsewhere diagnostic. The registration-attendance gap stays open because it never produces the loud, immediate signal this pathway answers. A live test is under way at Kadamba, where dinner sambar was dropped and a student complained it was removed ”while they are taking more money from us” (single-sourced); the change has not been reversed. Menu governance and registration governance have separate owners, so L4 proves the mechanism exists, not that it has ever been applied to the booking.

Candidate Loops

Each of these would close if its last link held, but that link is assumed, so each is drawn dashed and not counted as a loop.

L1: Speculative Registration / Resale Safety Net

A student registers without firm intent, producing an unused registration that feeds the Mess Cell resale market. Two respondents show the first links: ”Kadamba veg breakfast costs around Rs. 48 and I book it every day and never avail it,” and ”I have both bought and sold registrations.” Kadamba's manager describes the transfer from the kitchen side: students register ”even if you are in the village” and a friend eats in their place (single-sourced).

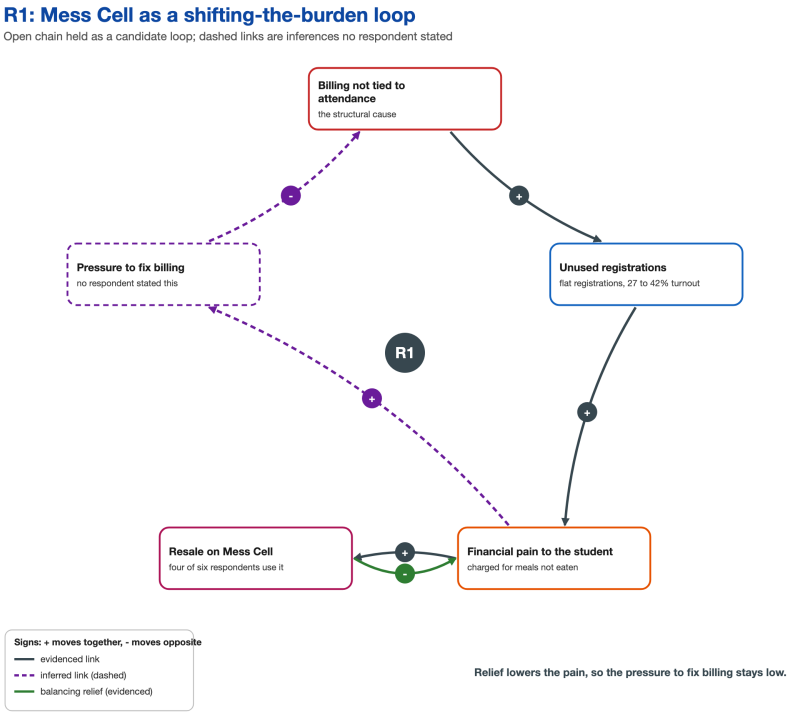


Figure 5. L1: Mess Cell as a shifting-the-burden loop; dashed links are inferences no respondent stated.

The first three links are confirmed. The closing link, that a lower net cost causes more speculative registration, is stated by no respondent, and the strongest student evidence points toward securing a scarce slot: ”I do register for breakfast for the whole month because Kadamba is hard to get and it gets full.” Registration also needs no motive, since it is automatic, so Kadamba's flat count of about 1,070 a day is what the default produces whatever anyone intends (L13). L1 is a behaviour on top of the default, not a driver of registration.

L9: Waste Absorption

Surplus passes through five routes before the bin. Yuktāhār reuses leftovers: idli becomes idli upma or idli poha for lunch, dosa batter goes back to the fridge and surplus milk is set as curd (confirmed). Kadamba cooks about 30 extra portions for 30 to 50 outsiders such as night-shift security, and about half its own staff eat from the same food (single-sourced); whether these portions sit inside or on top of the 70% first batch is unconfirmed. Food not eaten in the first round is held and reheated into later service (the team's judgement). Latecomers at the back door take a further share (L15). Kadamba's waste then goes to a dustbin emptied by a garbage contractor with no composting (single-sourced), though the CFS Chair describes an outside compost facility and the two accounts are unreconciled. At Bakul some leftovers return to Hafizpet and the rest is thrown.

The size of the absorption is where the evidence strains. Kadamba's operator puts waste at about 5 to 10% a day without saying of what (single-sourced). April averaged about 1,072 registered and 398 eaten a day (confirmed), so

a 70% first batch of about 750 leaves roughly 350 portions before any top-up (candidate arithmetic). By line, veg averaged 757 registered and 245 eaten, so a 70% batch is about 2.2 times what was eaten, and non-veg 315 and 153, about 1.4 times. A 5 to 10% figure fits only if outsiders, staff and held food absorb most of the surplus, if the batch is smaller in practice than the ratio implies, or if the figure uses another base. The evidence does not say which. Waste is visible to operators, per item in kilograms at Yuktāhār. What is invisible is the route from that record to anyone who sets the rules. The two links that would close the loop, that absorbed surplus suppresses pressure and that suppressed pressure permits more unused registration, have never been seen to fire. The better-grounded version is L17.

L17: Kitchen Competence Shifts the Burden

Operators discounting registrations and topping up live (L11, L12) relieve the gap's visible consequences every morning. Queues clear, shortages are re-cooked, and surplus is reused, fed to outsiders and staff, held or binned. Because each morning is handled locally, no signal accumulates upstream, which plausibly reduces pressure for the fundamental fix, a booking that tracks the daily decision to eat.

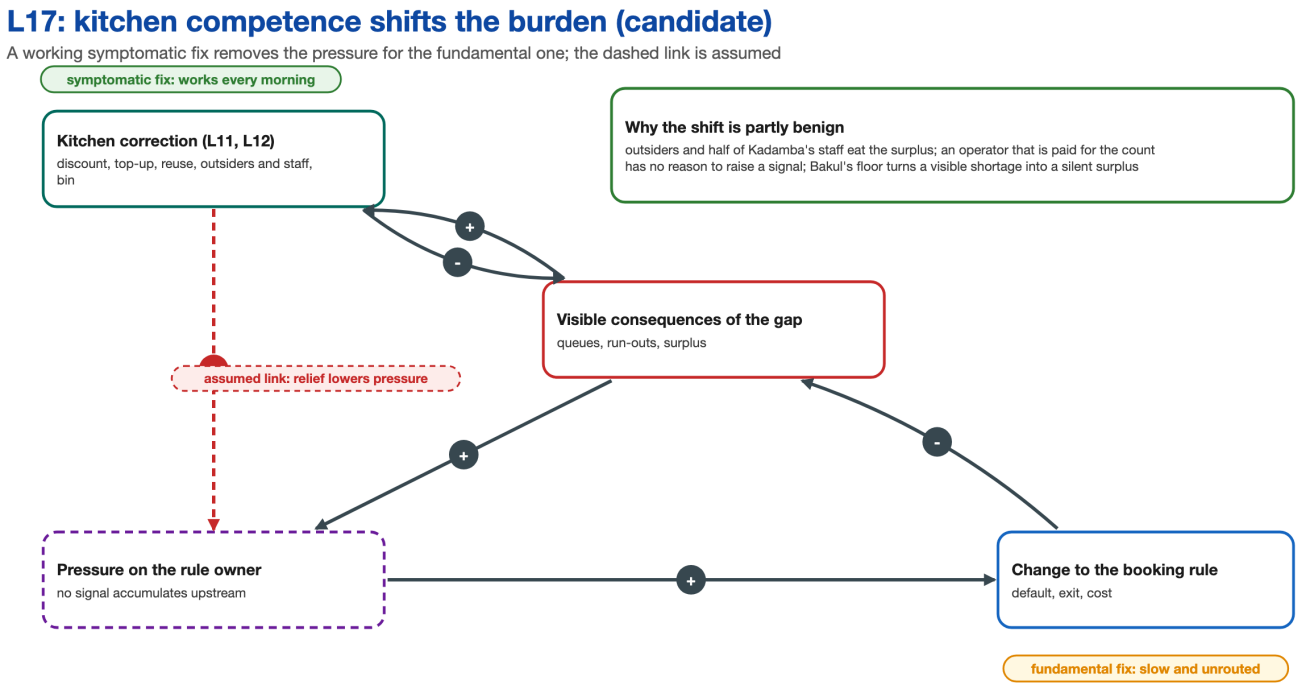


Figure 6. *L17: kitchen competence relieves the symptom each morning, so the fundamental fix never gets pressure.*

Two facts make the shift partly benign for those inside it. The surplus has beneficiaries, since night security and half of Kadamba's staff eat from the same food, so cutting the first batch takes their meals unless it plans for them. And an operator that treats the count as an obligation it is paid to meet has no motive to raise a signal upstream. Bakul's floor turns each low morning's risk of a visible shortage into a silent surplus, and if Bakul is paid on plates served, as its manager reports (L8), it bears that surplus itself and has accepted it (candidate). The symptomatic side is confirmed and the pressure-reduction link assumed, but it is better grounded than L9's version because the correction routes are observed. It is the clearest statement of why the system is stable and not broken.

L3b: Kadamba Demand Lock-in

Kadamba's reputation plausibly drives scarcity, and scarcity drives slot-locking for the respondent who registers all month because Kadamba fills up. April is consistent with that, but automatic registration predicts the same

pattern (about 1,070 a day, 62.9% unused, veg least used at 32.4%), so the aggregate supports the candidate less than it first appeared to. The Bakul operator's belief that Kadamba gets 90 to 95% turnout is contradicted for breakfast by the records. The closing link, that locking reinforces the reputation, is the weakest, and the only direct testimony describes a crowded Kadamba as a negative experience. It is retained as a plausible reason Kadamba stays saturated while Bakul stays under-registered, and is unconfirmed.

L15: Soft Close

At Kadamba, when the front door closes at 9:30 people come in by the back door and the mess cannot refuse them (single-sourced). The operator's wording covers dinner as well as breakfast, so breakfast back-door entry is a candidate, supported by the records: 9.2% of April scans came after 9:30, the latest at 10:07 (confirmed). If students learn the close is soft they arrive later, which builds the late crest and more latecomers, a cycle with no negative link. Only the first link is evidenced.

Causal Chains That Do Not Close

These are real, evidenced mechanisms, and none closes. For the first two that is the diagnostic point.

L13: Standing Booking to Fixed-Ratio Over-Preparation

This is the core chain of the Breakfast Paradox. Registration is mandatory and automatic, running whether or not the student is on campus. The only exit is five cancellations per meal type per month, each at least four days ahead, and beyond them the charge cannot be removed (confirmed). The registered count therefore stays flat across the week, 928 to 1,123 a day at Kadamba in April, about 1,070 on every weekday. Turnout did not. It ran from 42.1% on Tuesdays and 41.1% on Mondays to 34.8% on Saturdays and 27.3% on Sundays, and across the month 62.9% of bookings went unused. At Yuktāhār breakfast ran 17.7% to 40.2% while lunch under the same booking ran 56% to 89% (all confirmed). Whether the cycle is weekly, as Kadamba's operator describes it, or monthly, as students experience it, is unconfirmed.

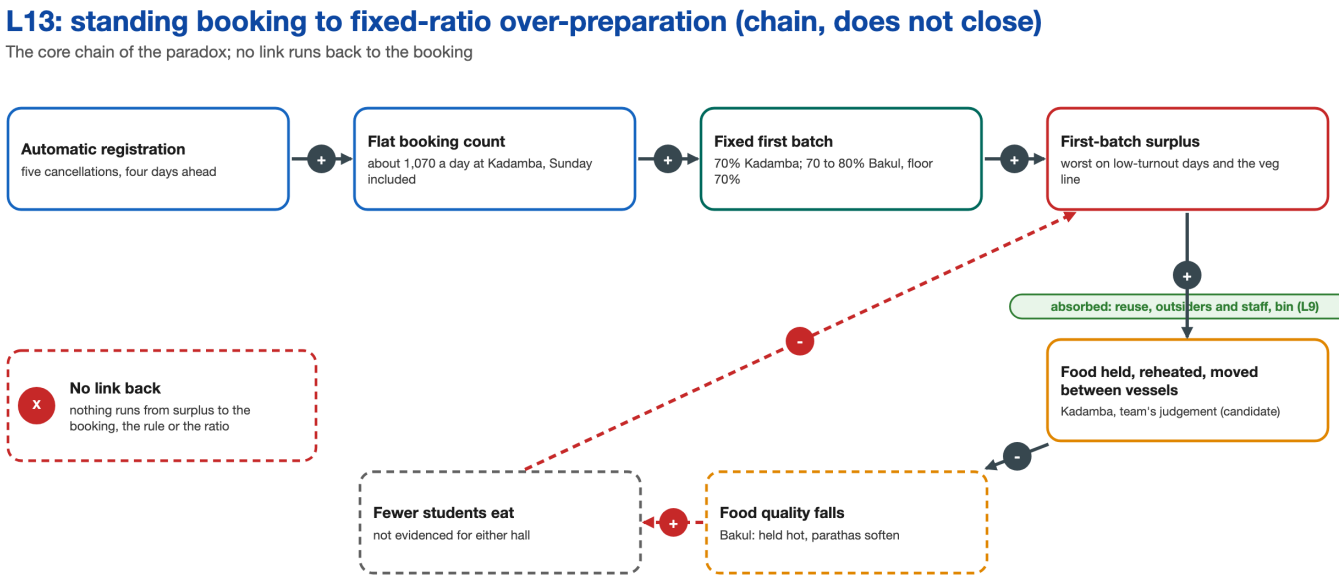


Figure 7. L13: the core chain. A flat booking meets a fixed ratio; the quality branch is dashed because its closing link is not evidenced.

The kitchen converts the flat count into a first batch under one of three regimes. Kadamba cooks a fixed 70%, veg and non-veg planned separately; Bakul cooks 70 to 80% by item and day, kept near 70% on low weekends; Yuktāhār uses no fixed ratio and plans from its books (L11). Kadamba and Bakul top up reactively. On Kadamba's numbers,

as candidate arithmetic, the first batch is about 1.9 times what April turnout needed, 2.6 on Sundays and 1.7 on Tuesdays, and about 2.2 times for veg against 1.4 for non-veg. A fixed ratio on a flat count over-prepares most on the days and the line where turnout is lowest. At Bakul the manager's own ceiling of about 60% sits below both his first batch and his floor, so over-preparation is built in on his numbers (candidate; no record to size it). Even full preparation runs short when portions drift: Kadamba's biryani lunch is cooked at 100%, two pieces of 150 to 200 g with unlimited rice, and when some diners take three, later ones go without. At breakfast bonda, puri, vada and idli ”float” above the gram standard (single-sourced).

Four delays separate the decision from the meal. A student who wants out must cancel at least four days ahead, which decouples the daily decision to eat from the bill. Quantity is set the day before, against procurement tiers: at Kadamba dry goods a week ahead, some vegetables two to three days ahead and perishables the day before (single-sourced). The chef arrives at 5:30, live items start at 7:00, and Bakul's labour-heavy items may start at Hafizpet around 3 AM for a 7 AM arrival. The first batch is thus fixed three and a half to four hours before the crest, after a food-safety sequence that must finish before any turnout signal exists (L8). No link runs from the surplus back to the booking or the rule. Operators control quantities, not rules, nothing routes their registers to the rule owner, and Kadamba treats the count as the obligation, so none runs back to the ratio either. That is the structural reason the gap persists: the only loops that act on it, L11 and L12, act on the kitchen's conversion and never on the booking. The same structure gives 56 to 89% at lunch, which is why the problem is breakfast's.

The surplus has one more consequence at Kadamba, an open branch. In the team's judgement (candidate), the main quality problem is food cooked ahead, reheated repeatedly and moved between vessels, one of them the display plate. More surplus means more holding and lower quality. If lower quality led to more skipping, lower turnout would raise the surplus and the branch would close into a reinforcing loop, but that link is not evidenced, so it is dashed. Bakul has a similar branch driven by distance: food cooked hours ahead and held hot softens, as hot parathas steam in the box (single-sourced, one item, operator only), and earlier dispatch or a bigger reserve lengthen the holding (candidate). Whether texture changes taste as students judge it, and whether taste moves turnout, is untested with any Bakul resident.

L14: Closing Rush to Just-in-Case Over-Preparation

Arrivals at Kadamba crest at close. In April 26.5% of scans fell between 9:15 and 9:30 and 13.5% between 9:00 and 9:15, the rate reached about nine a minute at 9:29, about 79 scans a day fell between 9:20 and 9:29, and the latest scan came at 10:07 (confirmed). Yuktāhār's site head says the rush is always at closing time (confirmed). Kadamba's rule of thumb, about 150 diners in the last ten minutes, matches the shape, and the records remain the measurement. Bakul's manager describes an earlier curve, a main rush at 8:00 to 8:30 with 10 to 15 in the last ten minutes (single-sourced; no scans), so curves differ by hall.

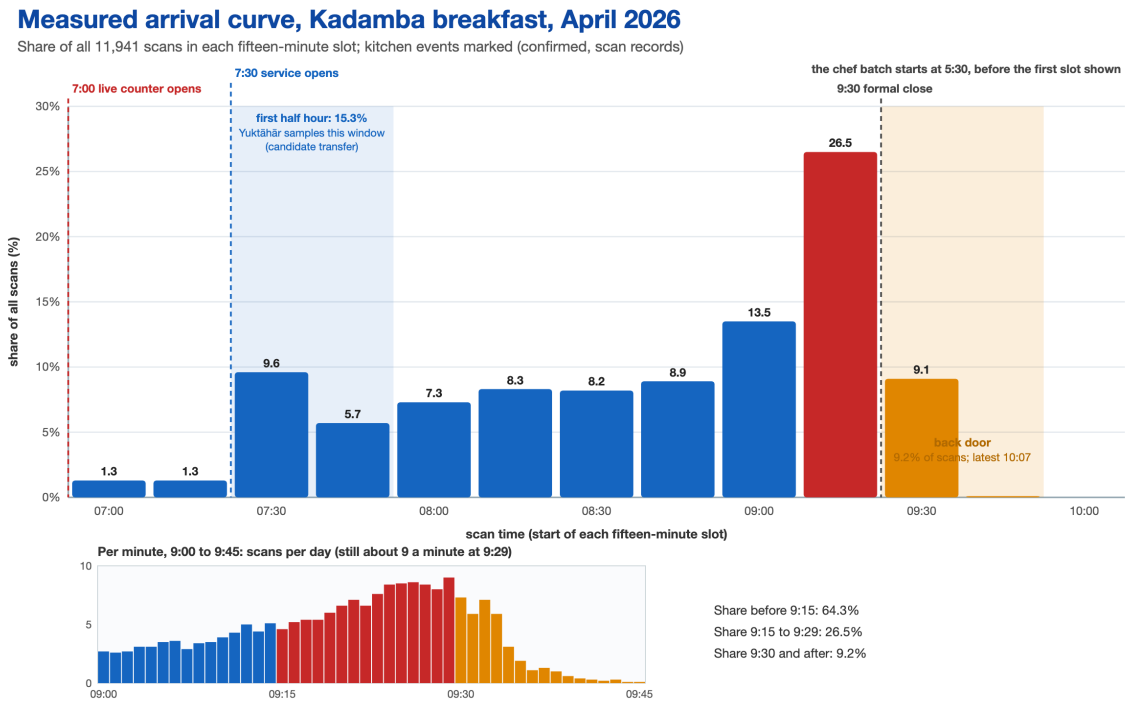


Figure 8. Kadamba breakfast arrivals, April 2026: the crest that a top-up cannot reach.

Where the crest comes at close, demand outruns what a top-up delivers in time, so one morning produces shortage and surplus together. Shortage shows as the 9:00 re-cook at Yuktāhār, back-door service at Kadamba and runs on the floating items. Surplus shows as Bakul's first batch and floor under its no-shortage rule, and Kadamba's raw reserve held for a quick cook. Kadamba's extra portions are not part of the hedge, since they feed outsiders as a planned commitment.

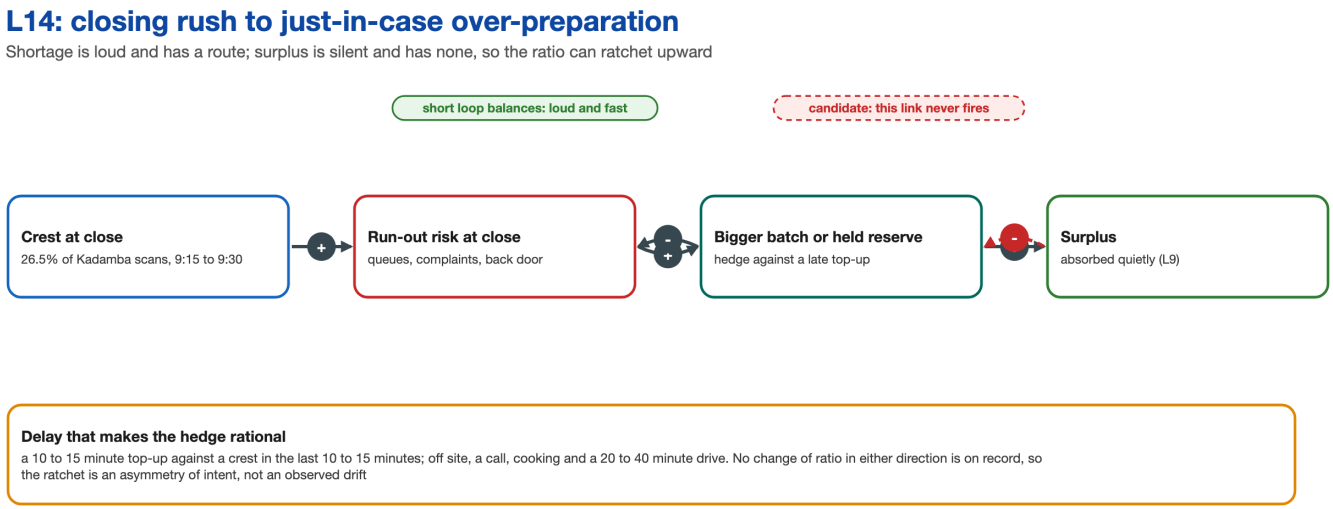


Figure 9. L14: shortage is loud and surplus is silent, so the ratio can ratchet upward. The dashed link is a candidate.

A candidate reinforcing closure sits on the chain. Shortages show as queues, complaints and back-door arrivals, while surplus is absorbed quietly. If an operator raises its ratio after run-outs and never lowers it after surplus, the ratio ratchets upward. No change of ratio in either direction is on record, so this is an asymmetry of intent, not an observed drift, and Bakul states its premise as a rule: shortage outweighs waste, and the weekly review stops at a

floor (single-sourced). Kadamba's operator also perceives the crest as larger than measured, about 43% of diners in the last ten minutes against about 20% on the records, a candidate reason a hedge would be set high. The chain matters without the ratchet. Part of the over-preparation is a rational hedge against a delay at close, not only a misreading of registration.

L2: Skip Meal: a Designed Mechanism That Does Not Close

Skip Meal lets a student who knows they will not attend tell the kitchen to prepare less. Its first link, use by students with real non-attendance intent, is confirmed and close to zero: four of six respondents who addressed it had never used it, one because ”it doesn't return the money.” Skip Meal tells the kitchen but does not remove the charge.

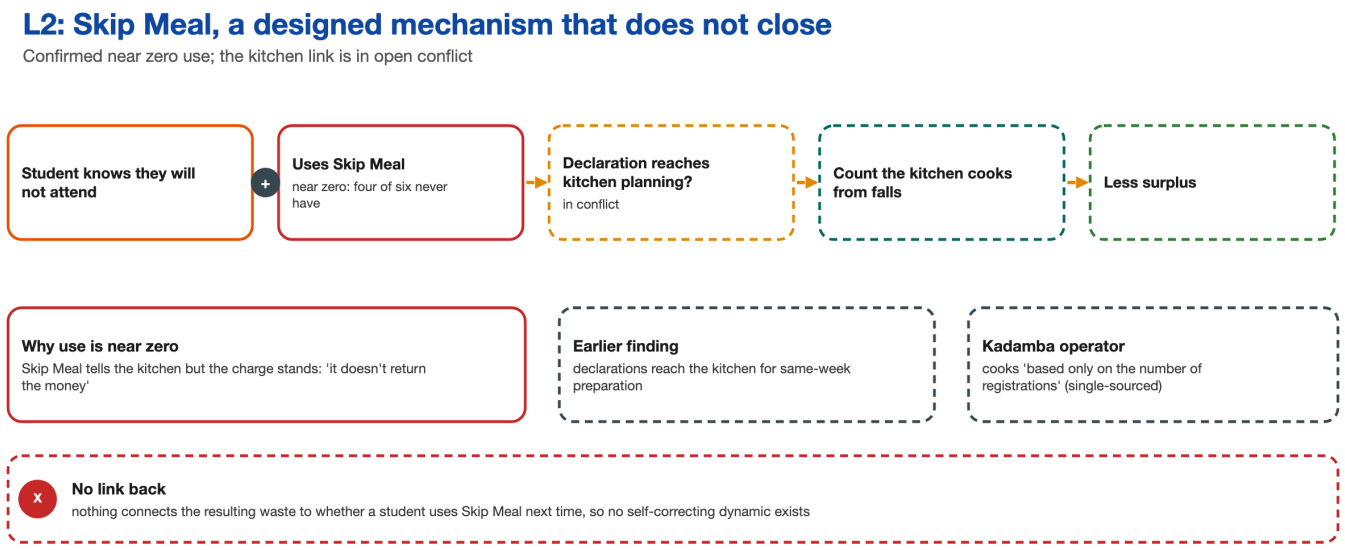


Figure 10. L2: Skip Meal. The student link is confirmed and near zero; the kitchen link is in open conflict.

The kitchen link is in open conflict. An earlier finding says declarations reach the kitchen for same-week preparation. All three operators plan from the portal registration count and none named Skip Meal as an input, and Kadamba's operator says the mess cooks ”based only on the number of registrations” and ”doesn't care if people don't want to eat” (single-sourced). The two fit together only if a declaration lowers the count the operator sees while the operator treats whatever count it sees as the obligation, which has not been checked. Even if the link holds, nothing connects the resulting waste back to whether a student uses Skip Meal next time, so no self-correcting dynamic exists to call a loop.

L5: Menu Transparency Effect

The rotation that closed L4 made the menu predictable, so a student can decide in advance not to come on a day they will dislike, a side effect the CFS Chair raised unprompted. Turnout does depend on the item: Yuktāhār's site head reports longer queues on aloo paratha and dosa days and 92% for Sunday-night paneer, Kadamba non-veg breakfasts ran at 48.5% against 32.4% for veg (both confirmed), and Bakul's manager says pav bhaji draws more students and muesli and Chocos go readily (single-sourced). Whether advance knowledge, rather than same-morning preference, causes skipping is untested. If the effect exists it flows forward into unused registrations on particular days, feeding L13.

L6: Shock-Adaptive Registration Relaxation

During an LPG shortage, and separately during some holidays and long weekends, the institution made registration non-compulsory. Students could walk in and scan, or register a day or two ahead, and were billed only for what they consumed. The trigger and the walk-in mechanism are confirmed, and Kadamba's operator describes cutting quantities and buying some items outside during the shock (single-sourced). The relaxation was episodic: students

say they stay auto-registered through ordinary holidays, so which holidays were relaxed is unconfirmed.

A first reading counted this as a self-limiting balancing loop in which relaxed billing raises procurement uncertainty and lowers institutional confidence in the model. Every link was assumed, and kitchens set cooked quantity the day before and buy dry goods that keep, so the first step is weak. The simpler reading fits better: the model reverts because the shortage or holiday ends, a calendar switch with no evidenced feedback. The candidate reasons it is never left on lie outside the kitchen, in whether per-plate payment counts registrations and in the convenience of an automatic default. Either way an attendance-tracked alternative has been built and has worked more than once. Yuktāhār's June Wastage Book pages add a candidate observation: two Saturday breakfasts with 37 and 34 registered record 33 and 31 diners, but those totals include unregistered and paying diners, and registered students alone numbered 22 and 21, about 60%, roughly double September's 28%. The regime in force in June is not known.

L7: Late-Night Workload to Skip to Crash

Staying up late is the best-evidenced single cause of skipping in the project: ”If I sleep around 3:00 or 4:00 a.m. or later, I don't tend to wake up. I prefer sleep over breakfast.” The records match. Sunday is Kadamba's lowest breakfast at 27.3% and among Yuktāhār's lowest at 17.9% to 24.1%, and Kadamba's median scan falls between 8:51 and 9:11 with the crest at close, consistent with late waking (confirmed, as a population pattern, since the export carries no student identifier). Yuktāhār's site head names hackathons and exams as reasons numbers fall. Bakul's manager gives two reasons, not waking and taste or menu. The first agrees with students, and the second is his hypothesis, which the six interviews rarely support and none of whom is known to live at Bakul. What follows a skipped breakfast, an energy crash, a compensating purchase or a change in that night's sleep, has never been asked, so the rest of the chain is untested. Late waking is also a candidate contributor to the closing crest.

Absent and Disconfirmed Mechanisms

A missing or disconfirmed loop is as much a finding as a working one.

L3, capacity redistribution, is strongly contraindicated. The plausible story, that scarcity at Kadamba pushes students to Bakul or Palash, runs against the evidence: scarcity makes students lock Kadamba down harder, as the respondent who registers all month says directly. Bakul's low registration is better explained by a standing preference gap. This rests on one quote corroborated by the aggregate pattern, hence ”strongly contraindicated” and not disconfirmed, and Bakul and Palash have no attendance records.

L8, vendor quality discipline, is unresolved. Student billing tracks registration (confirmed). The vendor's payment is a different link and the evidence on it conflicts. Bakul's manager says payment is a monthly lump sum with deductions, based on plates served, but ”served” is the synthesis's wording of one remark and may mean plates billed. Kadamba's ”they are getting paid for it” is a hint that payment follows registration, not a confirmation. Contracts may differ by operator and none has been seen. If vendors are paid per registered plate, falling attendance never touches vendor revenue and a fixed ratio turns part of the gap into margin. If per served plate, the vendor has a direct stake in turnout and the loop could close. Yuktāhār shows a third route: a vendor that ties waste reduction to profit gains from cutting waste whatever the plate basis. Operators already respond to item-level signals. Kadamba cut ragi idli and raised plain idli, Bakul switched paneer supplier twice after complaints, and Yuktāhār changes supplier when a grams-of-debris norm is exceeded (all single-sourced).

Quality control before service is a working balancing structure at Kadamba, on taste and safety, not quantity. Each meal has its cooking temperature checked, a sample kept for 72 hours, a wrapped display plate set at the entrance, serving temperature measured per item, and a tasting by the facility team with two Prism staff, then corrections (display plate and tasting confirmed as observed, the rest single-sourced). QC inspects weekly and water TDS is read daily. Nothing in the sequence tests first-batch size. Complaints run down a chain from CFS to CDS to Prism, an operations manager, a manager, a supervisor and the service staff, by director's email into a WhatsApp group at Kadamba and from the central dining team to the manager, the kitchen and the managing director at Bakul. Two different people are on record as head of CDS, and whether one name is wrong or they hold two roles is open.

L16, operator waste to menu choice, is absent. Operators see which items waste: at Kadamba uttapam, set dosa and upma consistently, while puri, bonda, vada and idli move (single-sourced). All three say students or the

committee decide the menu, and Bakul's manager is asking for input. He holds knowledge the menu process does not use, which items go (pav bhaji, muesli, Chocos) and which pairings work: kachori with chole, and Raj Kachori with aloo curry rather than the chana it is listed with, from about 20 years with North Indian food. That is his view of the students' reasoning, not theirs, and whether Bakul residents share it is untested. Decision rights sever the loop, since operators hold the data and the committee holds the choice. It is the counterpart of L4, which closes because complaints are loud and have a route while waste is silent and has none.

L10, the measured-but-unrouted gap, is a static condition and not a spiral. The gap is measured in at least three places: the CDS scan export (62.9% no-shows for Kadamba breakfast), Yuktāhār's register (roughly 72% of registered breakfasts unused), and Yuktāhār's other books, which are a Wastage Book, a wastage board, a grocery issue book, a store and ingredient book, a stock register, a vegetable book and indent, a milk, fruit, gas, electricity and water register, and fridge logs (confirmed). At Bakul the measurement is the manager's weekly review and may be unwritten. The CFS Chair's 35 to 40% breakfast figure is now confirmed for Kadamba at 37.1%. Predictable skipping is noticed and acted on in the quantity of food and never in the registration rule. The failure is a missing route, not a missing instrument, and the instrument is hard to route even where it exists, since Yuktāhār's records are on paper, digitised later by office staff. The chain from CFS through CDS to Prism runs downward, and whether any operator data runs back up it is unknown. Operators are data-rich and authority-poor: even information they need reaches them informally, as the site head says of events and exams, ”We don't come to know... until we sit in the meeting.” The condition stays static because the kitchen loops absorb each morning's consequence locally, and nothing tracks the gap's visibility to decision-makers over time, so there is no basis for calling it reinforcing.

Unintended Consequences

Several mechanisms are side effects of fixes introduced for other reasons, and two run in a helpful direction.

Rule or action	Intended effect	Unintended consequence	Status
Biweekly menu rotation (L4)	End menu fatigue	Students can plan to skip disliked days (L5); the matching-day Wastage Book lookup becomes possible (L11)	Lookup confirmed; skipping untested
Automatic registration, exit four days ahead	Procurement certainty	A default that runs while students are away (L13), feeding resale (L1); perishables are ordered the day before anyway, so the lock buys little	Rule confirmed; tiers single-sourced
Back-door entry after 9:30	Serve latecomers	May teach students the close is soft (L15)	Practice single-sourced; consequence candidate
Bakul's shortage floor	No shortage	More surplus on every low morning; a review that cannot correct below the floor (L11, L14)	Rule single-sourced; consequence candidate
Off-site cooking and hot holding (Bakul)	Hot, safe food after 8 km	Parathas soften; the late reserve is the food held longest (L12, L13)	One item single-sourced; consequence candidate
Fixed 70% batch with reactive top-up (Kadamba)	Avoid shortage	Surplus held, reheated and moved between vessels (L13)	Candidate
Food-safety sequence	Safety and taste	Fixes the first batch hours before any turnout signal	Plate and tasting confirmed
About 30 extra portions for outsiders; staff eat from the same food	Feed night security and staff	Surplus has beneficiaries, so calibrating the batch has losers (L9, L17)	Single-sourced; consequence candidate
Waste tracking at Yuktāhār	Reduce waste	Staff resistance that became ownership once weighing was blame-free	Self-report
Trying and dropping a paid ERP	Stay close to own data	Learning stays on paper, hard to route beyond the kitchen (L10)	Single-sourced
Skip Meal with no refund	Let students signal absence	Near-zero use, so the kitchen gets no signal (L2)	Confirmed

Structural Causes and Systemic Tensions

The three structural causes from the systemic problem analysis explain much of what closes and what does not, and one intermediate mechanism explains the kitchen half.

Registration is an automatic, mandatory standing booking with an exit by five early cancellations. It cannot follow the morning decision to eat, which keeps the count flat in L13 whatever the motive behind each booking, and which is why the problem is breakfast's. Student billing fires at the booking, not at consumption. That allows L1's cheap speculation, underlies L3b and L9, and is what L6 shows the institution can suspend; whether vendor payment follows the same rule is open (L8). The gap is measured but unrouted, because decision rights over menu, kitchen quantity, academic scheduling, registration and billing sit with different actors and none holds more than one or two. That fragmentation is why L4 can close, since menu has a real owner and an escalation path, while L10 and L16 cannot.

Beneath the kitchen half, three kitchens convert registrations into a first batch under three regimes, and every kitchen can correct only upward within a meal. Kadamba applies a fixed ratio with no learning record, and its gram standards fix portions, not quantities. Bakul adjusts weekly by judgement down to a floor and tops up by phone from 20 to 40 minutes away. Yuktāhār calibrates from its books and projects from the first half hour, yet anchors on a wrong belief that turnout is about half. This sits downstream of the three causes, inside authority the operator already holds. It explains why L13 produces surplus on low-turnout days, why L12 cannot remove it, and why L14's hedge can ratchet.

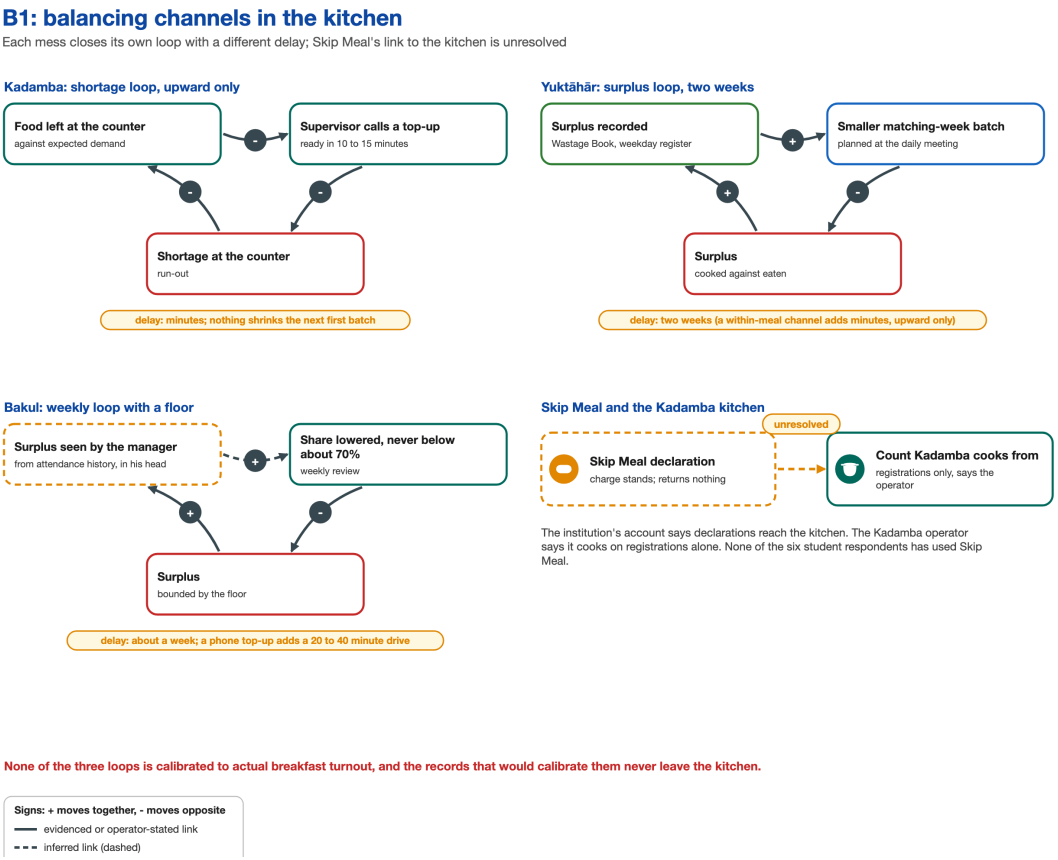


Figure 11. *The three kitchens close different loops with different delays.*

Delays set the timing. Raw surplus is largely recoverable, since Yuktāhār buys groceries fortnightly, vegetables two or three times a week and milk daily (confirmed), while cooked surplus is not. The binding delay is between the cooked first batch and the late crest, not the four-day cancellation lead. Yuktāhār's learning loop runs on a lag of one to two weeks and its culture change took about four months. The rest do not close for narrower reasons: L2 because Skip Meal pays the student nothing, L5 because item-level turnout is noticed only in the kitchen, L7 because nobody has asked what follows a skip. The evidence runs out at three points: the plate basis that decides L8, the absence of records for Bakul, Palash and for lunch and dinner at Kadamba, and the base of Kadamba's 5 to 10% waste figure.

Eight tensions follow, in three groups. Between timing and certainty: the four-day lead pulls against perishables ordered the day before, and operators themselves challenged the cut-off; avoiding shortage at close pulls against

avoiding waste after it (L14), a trade Bakul settles by rule for no shortage. Between quality and batch cooking: aloo paratha is best eaten at once yet may start around 3 AM for a 7 AM arrival, so holding keeps temperature and loses texture; Yuktāhār's head argues against batching because ”student fluctuation is very high” while cooked food lasts three to four hours; and at Kadamba food is held, reheated and transferred until it degrades (L13). Between who holds what: operators hold the data while others hold the authority (L10, L16), cutting waste pulls against feeding the people who eat the surplus (L9, L17), and one registration rule leads one operator to cook to the count as a paid obligation, another to learn waste away as lost profit and a third to accept waste as the price of never running short.

What This Means for the Reframe

Two kinds of closed loop are now demonstrated. L4 shows governance closing a loop end to end when a complaint is loud enough, and L11 and L12 show the kitchen closing loops every day, within the meal and across weeks. Neither touches registration. L1 and L6 fit a stable equilibrium but neither closes on the evidence, since L1's closing link is assumed and L6 is a calendar switch.

The steadier support for the equilibrium reading sits in two places. The first is the absorption layer: resale and informal transfer, the operator's discount and live top-up, reuse at Yuktāhār, outside diners and staff at Kadamba, held and reheated food (candidate), back-door service, outside venues, and finally the bin. The second is the asymmetry of attention. Shortages are loud and trigger fast correction, while surplus is silent, can ratchet the ratio upward (L14) and never reaches the menu owner (L16). As a candidate reading, Bakul's floor is the kitchen-level form of the same attention to what is loud, and the kitchen's competence at absorbing the gap is the plausible reason no signal accumulates upstream (L17). Under the same booking rule, operator culture sets how much of the gap becomes surplus, so the kitchen can learn, locally, and at Bakul only down to a floor. A precedent for solving the core problem exists in L6, and two designed corrective mechanisms do not connect back to the behaviour they were built to correct: L2, and L8 if vendors are paid per registered plate. Read together, the evidence supports organisational inertia around a problem that is measured, locally managed and previously solved at least as strongly as an actively self-reinforcing spiral.

What Remains Untested

Several links can be settled with one question or one record.

- The plate basis per operator, from the contract or CDS, which decides L8 and whether a fixed ratio turns the gap into vendor margin.
- Whether a Skip Meal declaration lowers the count Kadamba cooks against (L2).
- Whether any operator has ever changed a standing ratio, up after run-outs or down after surplus, which tests the L14 ratchet.
- The base of Kadamba's 5 to 10% waste figure and whether a record exists; whether the roughly 30 outside portions sit inside or on top of the 70% batch; how many people ”half the staff” is; and whether staff meals are an entitlement or a norm.
- Whether held surplus at Kadamba is reheated into later service, and whether reheated or softened food drives complaints or skipping, including whether holding changes how Bakul residents judge the food (L13's quality branches).
- Whether Yuktāhār uses its first-half-hour projection at breakfast and adjusts for the crest, whether its issue quantities fall over successive matching weeks, and which of its two 3 September diner records is right.
- Which registration regime applied in Yuktāhār's June breakfasts and which holidays were relaxed (L6); whether the cycle is weekly or monthly; and who heads CDS.
- Whether back-door entry happens at breakfast and shifts arrival later (L15).
- Whether Bakul's weekly review writes anything down, the real Hafizpet top-up lead time, whether the 9:20 reserve rests on data or judgement, Bakul breakfast scans to test the 60% estimate and the 8:00 to 8:30 rush, whether Palash shares the kitchen, and where Yuktāhār's unreused surplus finally goes.
- What follows a skipped breakfast (L7), whether advance knowledge of the menu causes skipping (L5), and why

Bakul residents skip, asked of them and not inferred from the manager.